## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Simplify 2 $^{3} \times 2^{4}$.

2. Express 3 $^{-2}$.

3. Solve 2 $x + 1 = 5$.

4. Differentiate $x^2 + 2x$.

5. A decrease of 20% gives which multiplier?

6. For a geometric sequence $u_0 = 5, q = 2$, calculate $u_n$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** Explain the difference between a continuous rate and a discrete multiplier; present an inequality proved via the minimum of a difference.

## Investigate before formalising

Compare the increments of $e^x$ and the values of the function around 0, with several steps positive. Carry out three Euler steps and explain the rule chosen.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.